Hadoop Assignment 01 - Report

Big Data Processing - MapReduce Jobs

Team Information

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Assignment Overview

This assignment demonstrates proficiency in:

1. Creating synthetic big data datasets
2. Working with HDFS (Hadoop Distributed File System)
3. Writing MapReduce jobs with Mapper and Reducer functions
4. Implementing Combiners for optimization

Step 1: Dataset Generation (2 Marks)

Implementation Details

Program: `DatasetGenerator.java`

Datasets Created:

1. **Buyers.txt** - 10,000 records
   - BuyerID: Sequential (1-10,000)
   - BuyerName: Random string (10-15 chars)
   - BuyerAge: Random (12-75)
   - BuyerGender: Random (male/female)
   - BuyerSalary: Random float (3500-11000)
2. **Purchases.txt** - 1,000,000 records
   - purchID: Sequential (1-1,000,000)
   - BuyerID: Random reference (1-10,000)
   - purchPrice: Random float (10-100)
   - purchNumItems: Random integer (1-10)

Compilation and Execution Commands

```
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ javac DatasetGenerator.java && java DatasetGenerator
Generating Buyers dataset...
Generated 1000 buyers...
Generated 2000 buyers...
Generated 3000 buyers...
Generated 4000 buyers...
Generated 5000 buyers...
Generated 6000 buyers...
Generated 7000 buyers...
Generated 8000 buyers...
Generated 9000 buyers...
Generated 10000 buyers...
Buyers dataset created successfully!
Generating Purchases dataset...
Generated 50000 purchases...
Generated 100000 purchases...
Generated 150000 purchases...
Generated 200000 purchases...
Generated 250000 purchases...
Generated 300000 purchases...
Generated 350000 purchases...
Generated 400000 purchases...
Generated 450000 purchases...
Generated 500000 purchases...
Generated 550000 purchases...
Generated 600000 purchases...
Generated 650000 purchases...
Generated 700000 purchases...
Generated 750000 purchases...
Generated 800000 purchases...
Generated 850000 purchases...
Generated 900000 purchases...
Generated 950000 purchases...
Generated 1000000 purchases...
Purchases dataset created successfully!

Dataset generation completed!
Files created: Buyers.txt and Purchases.txt
```

Figure 1: Terminal output showing successful generation

```
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ ls -lh Buyers.txt Purchases.txt
-rw-rw-r-- 1 radeon radeon 347K Oct 11 15:49 Buyers.txt
-rw-rw-r-- 1 radeon radeon  19M Oct 11 15:49 Purchases.txt
```

Figure 2: File sizes using `ls -lh Buyers.txt Purchases.txt`

```
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ head -10 Buyers.txt
1,6Y9jo5SnIg,14,male,9995.01
2,DlC2OmAANbj,34,female,4975.89
3,MS3DCoHWIYdol8z,53,female,6261.16
4,79SMsUJshBzcuv,51,female,7376.80
5,5aCLa0OI2zZ,16,female,5450.10
6,Ub8v5IwWkzt,59,male,6602.51
7,rNpTqtMInVg,22,female,10895.79
8,WOl328SSEpg1K,69,male,9294.93
9,hrShVzO3fIHbL,40,male,9721.95
10,k6OhjxHYS0XgeI,14,female,9242.12
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$
```

Figure 3: First 10 lines of each file using `head -10 Buyers.txt`

Step 2: Upload to HDFS (2 Marks)

Commands Used

 # Create directory in HDFS

1. hdfs dfs -mkdir -p /user/hadoop/input

 # Upload Buyers dataset

2. hdfs dfs -put Buyers.txt /user/hadoop/input/

 # Upload Purchases dataset

3. hdfs dfs -put Purchases.txt /user/hadoop/input/

```
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ hdfs dfs -mkdir -p /user/hadoop/input
25/10/11 15:50:08 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-ja
va classes where applicable
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ hdfs dfs -put Buyers.txt Purchases.txt /user/hadoop/input/
hdfs dfs -ls /user/hadoop/input/
25/10/11 15:51:47 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-ja
va classes where applicable
25/10/11 15:51:49 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-ja
va classes where applicable
Found 2 items
-rw-r--r--   3 radeon supergroup     355177 2025-10-11 15:51 /user/hadoop/input/Buyers.txt
-rw-r--r--   3 radeon supergroup   19878385 2025-10-11 15:51 /user/hadoop/input/Purchases.txt
```

Figure 4: Executing commands 1, 2, and 3 of step 2

 # Verify upload

hdfs dfs -ls /user/hadoop/input/

```
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ # Verify upload
hdfs dfs -ls /user/hadoop/input/
25/10/11 16:49:21 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
-rw-r--r--   3 radeon supergroup     355177 2025-10-11 15:51 /user/hadoop/input/Buyers.txt
-rw-r--r--   3 radeon supergroup   19878385 2025-10-11 15:51 /user/hadoop/input/Purchases.txt
```

Figure 5: Executing command 4 Verifying upload

 # Check file details

hdfs dfs -du -h /user/hadoop/input/

```
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ hdfs dfs -du -h /user/hadoop/input/
25/10/11 16:18:32 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-ja
va classes where applicable
346.9 K  /user/hadoop/input/Buyers.txt
19.0 M   /user/hadoop/input/Purchases.txt
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$
```

Figure 6: Checking file details

Verification Steps

Check via Command Line:

```
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ hdfs dfs -ls -h /user/hadoop/input/
25/10/11 16:20:15 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-ja
va classes where applicable
Found 2 items
-rw-r--r--   3 radeon supergroup    346.9 K 2025-10-11 15:51 /user/hadoop/input/Buyers.txt
-rw-r--r--   3 radeon supergroup     19.0 M 2025-10-11 15:51 /user/hadoop/input/Purchases.txt
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$
```

Figure 7: Terminal output of `hdfs dfs -ls` command

```
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ hdfs dfs -cat /user/hadoop/input/Buyers.txt | head -5
25/10/11 16:20:51 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-ja
va classes where applicable
1,6Y9jo5SnIg,14,male,9995.01
2,DlC2OmAANbj,34,female,4975.89
3,MS3DCoHWIYdol8z,53,female,6261.16
4,79SMsUJshBzcuv,51,female,7376.80
5,5aCLa0OI2zZ,16,female,5450.10
cat: Unable to write to output stream.
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$
```

Figure 8:First 5 lines of Buyers.txt hdfs dfs -cat /user/hadoop/input/Buyers.txt | head -5

Check via Web UI:

- Open browser: `http://localhost:9870`
- Navigate to: Utilities → Browse the file system
- Navigate to: `/user/hadoop/input/`

Step 3: MapReduce Jobs

Job 1: Age Filter (20-50 years) - 4 Marks

Job Analysis

Type: Map-Only Job (no aggregation needed)

Mapper Output:

- Key: BuyerID (Text)
- Value: BuyerName, Age, Gender, Salary (Text)

Logic: Filter records where age >= 20 AND age <= 50

Implementation: `AgeFilterJob.java`

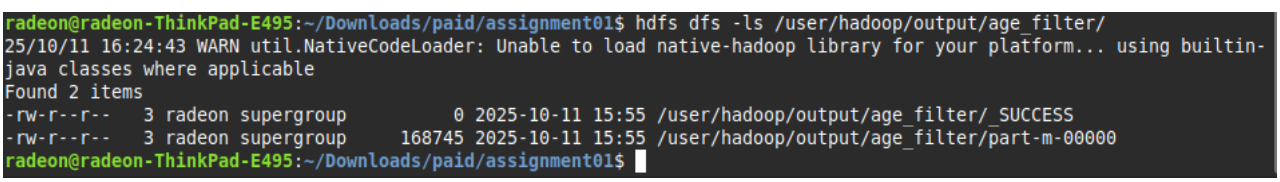
Key Features:

- Parses CSV input
- Filters based on age condition
- No reducer needed (setNumReduceTasks(0))
- Outputs complete buyer information

Compilation and Execution

bash

```bash
# Set Hadoop classpath
export HADOOP_CLASSPATH=$(hadoop classpath)

# Compile the job
javac -classpath $HADOOP_CLASSPATH AgeFilterJob.java

# Create JAR file
jar -cvf agefilter.jar *.class

# Remove old output directory if exists
hdfs dfs -rm -r /user/hadoop/output/age_filter

# Run the MapReduce job
hadoop jar agefilter.jar AgeFilterJob /user/hadoop/input/Buyers.txt /user/hadoop/output/age_filter

# View results
hdfs dfs -cat /user/hadoop/output/age_filter/part-m-* | head -20
```



```
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ hdfs dfs -ls /user/hadoop/output/age_filter/
25/10/11 16:24:43 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-
java classes where applicable
Found 2 items
-rw-r--r--   3 radeon supergroup          0 2025-10-11 15:55 /user/hadoop/output/age_filter/_SUCCESS
-rw-r--r--   3 radeon supergroup     168745 2025-10-11 15:55 /user/hadoop/output/age_filter/part-m-00000
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$
```

Figure 9: Output file in HDFS: `hdfs dfs -ls/user/hadoop/output/age_filter/`

Figure 10: Sample output records using `hdfs dfs -cat` command

```
-rw-r--r--   3 radeon supergroup     168743 2025-10-11 15:55 /user/hadoop/output/age_filter/part-m-00000
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ hdfs dfs -cat
-cat: Not enough arguments: expected 1 but got 0
Usage: hadoop fs [generic options]
        [-appendToFile <localsrc> ... <dst>]
        [-cat [-ignoreCrc] <src> ...]
        [-checksum <src> ...]
        [-chgrp [-R] GROUP PATH...]
        [-chmod [-R] <MODE[,MODE]... | OCTALMODE> PATH...]
        [-chown [-R] [OWNER][:[GROUP]] PATH...]
        [-copyFromLocal [-f] [-p] [-l] [-d] <localsrc> ... <dst>]
        [-copyToLocal [-f] [-p] [-ignoreCrc] [-crc] <src> ... <localdst>]
        [-count [-q] [-h] [-v] [-t [<storage type>]] [-u] [-x] <path> ...]
        [-cp [-f] [-p | -p[topax]] [-d] <src> ... <dst>]
        [-createSnapshot <snapshotDir> [<snapshotName>]]
        [-deleteSnapshot <snapshotDir> <snapshotName>]
        [-df [-h] [<path> ...]]
        [-du [-s] [-h] [-x] <path> ...]
        [-expunge]
        [-find <path> ... <expression> ...]
        [-get [-f] [-p] [-ignoreCrc] [-crc] <src> ... <localdst>]
        [-getfacl [-R] <path>]
        [-getfattr [-R] {-n name | -d} [-e en] <path>]
        [-getmerge [-nl] [-skip-empty-file] <src> <localdst>]
        [-help [cmd ...]]
        [-ls [-C] [-d] [-h] [-q] [-R] [-t] [-S] [-r] [-u] [<path> ...]]
        [-mkdir [-p] <path> ...]
        [-moveFromLocal <localsrc> ... <dst>]
        [-moveToLocal <src> <localdst>]
        [-mv <src> ... <dst>]
        [-put [-f] [-p] [-l] [-d] <localsrc> ... <dst>]
        [-renameSnapshot <snapshotDir> <oldName> <newName>]
        [-rm [-f] [-r|-R] [-skipTrash] [-safely] <src> ...]
        [-rmdir [--ignore-fail-on-non-empty] <dir> ...]
        [-setfacl [-R] [{-b|-k} {-m|-x <acl_spec>} <path>]|[--set <acl_spec> <path>]]
        [-setfattr {-n name [-v value] | -x name} <path>]
        [-setrep [-R] [-w] <rep> <path> ...]
        [-stat [format] <path> ...]
        [-tail [-f] <file>]
        [-test -[defsz] <path>]
        [-text [-ignoreCrc] <src> ...]
        [-touchz <path> ...]
        [-truncate [-w] <length> <path> ...]
        [-usage [cmd ...]]

Generic options supported are:
-conf <configuration file>        specify an application configuration file
-D <property=value>               define a value for a given property
-fs <file:///|hdfs://namenode:port> specify default filesystem URL to use, overrides 'fs.defaultFS' property from configurations.
-jt <local|resourcemanager:port>  specify a ResourceManager
-files <file1,...>                specify a comma-separated list of files to be copied to the map reduce cluster
-libjars <jar1,...>               specify a comma-separated list of jar files to be included in the classpath
-archives <archive1,...>          specify a comma-separated list of archives to be unarchived on the compute machines

The general command line syntax is:
command [genericOptions] [commandOptions]

Usage: hadoop fs [generic options] -cat [-ignoreCrc] <src> ...
```

Job 2: Purchase Statistics with Combiner

Job Analysis

Type: Full MapReduce Job with Combiner

Mapper Output:

- Key: BuyerID (IntWritable)
- Value: CountSumWritable (count=1, sum=purchPrice)

Combiner Function:

- Aggregates partial results from mappers
- Reduces network traffic between mappers and reducers
- Same logic as reducer (associative and commutative)

Reducer Output:

- Key: BuyerID (IntWritable)
- Value: "numPurchases,sumPrices" (Text)

Implementation: `PurchaseStatsJob.java`

Key Features:

- Custom Writable class (CountSumWritable) for count and sum
- Mapper emits (BuyerID, (1, price)) for each purchase
- **Combiner aggregates partial results (0.5 marks)**
- Reducer produces final aggregation per buyer

Compilation and Execution

bash

```
# Compile all classes including custom Writable
1. javac -classpath $HADOOP_CLASSPATH PurchaseStatsJob.java

# Create JAR file
2. jar -cvf purchasestats.jar *.class
```

```
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ javac -classpath $HADOOP_CLASSPATH PurchaseStatsJob.java
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ jar -cvf purchasestats.jar *.class
added manifest
adding: AgeFilterJob$AgeFilterMapper.class(in = 2104) (out= 954)(deflated 54%)
adding: AgeFilterJob.class(in = 1545) (out= 862)(deflated 44%)
adding: PurchaseStatsJob$CountSumWritable.class(in = 1946) (out= 951)(deflated 51%)
adding: PurchaseStatsJob$PurchaseCombiner.class(in = 1859) (out= 779)(deflated 58%)
adding: PurchaseStatsJob$PurchaseMapper.class(in = 2568) (out= 1109)(deflated 56%)
adding: PurchaseStatsJob$PurchaseReducer.class(in = 2540) (out= 1068)(deflated 57%)
adding: PurchaseStatsJob.class(in = 1964) (out= 1047)(deflated 46%)
```

Figure 11: Executing commands 1, 2

```
# Remove old output directory if exists
3. hdfs dfs -rm -r /user/hadoop/output/purchase_stats

# Run the MapReduce job
4. hadoop jar purchasestats.jar PurchaseStatsJob /user/hadoop/input/Purchases.txt /user/hadoop/output/purchase_stats
```

```
adding: PurchaseStatsJob.class(in = 1964) (out= 1047)(deflated 46%)
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ hadoop jar purchasestats.jar PurchaseStatsJob /user/hadoop/input/Purchases.txt /user/hadoop/output/purchase_stats
Exception in thread "main" java.lang.UnsupportedClassVersionError: PurchaseStatsJob has been compiled by a more recent version of the Java Runtime (class file version 65.0), this version
of the Java Runtime only recognizes class file versions up to 52.0
        at java.lang.ClassLoader.defineClass1(Native Method)
        at java.lang.ClassLoader.defineClass(ClassLoader.java:756)
        at java.security.SecureClassLoader.defineClass(SecureClassLoader.java:142)
        at java.net.URLClassLoader.defineClass(URLClassLoader.java:473)
        at java.net.URLClassLoader.access$100(URLClassLoader.java:74)
        at java.net.URLClassLoader$1.run(URLClassLoader.java:369)
        at java.net.URLClassLoader$1.run(URLClassLoader.java:363)
        at java.security.AccessController.doPrivileged(Native Method)
        at java.net.URLClassLoader.findClass(URLClassLoader.java:362)
        at java.lang.ClassLoader.loadClass(ClassLoader.java:418)
        at java.lang.ClassLoader.loadClass(ClassLoader.java:351)
        at java.lang.Class.forName0(Native Method)
        at java.lang.Class.forName(Class.java:348)
        at org.apache.hadoop.util.RunJar.run(RunJar.java:232)
        at org.apache.hadoop.util.RunJar.main(RunJar.java:153)
```

Figure 12: MapReduce job

```
# View results
4. hdfs dfs -cat /user/hadoop/output/purchase_stats/part-r-* | head -20
```

```
cat: Unable to write to output stream.
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ hdfs dfs -cat /user/hadoop/output/purchase_stats/part-r-* | sort -t',' -k2 -nr | head -10
25/10/11 16:28:14 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
5732    132,8029.48
3331    134,7747.84
2966    134,7709.36
3219    132,7702.05
1738    130,7665.90
4400    139,7649.50
906     127,7632.09
7517    133,7570.08
```

Figure 13: Results of 20 lines

```
# Sort by number of purchases (descending)

hdfs dfs -cat /user/hadoop/output/purchase_stats/part-r-* | sort -t',' -k2 -nr | head -10
```

```
        at org.apache.hadoop.util.RunJar.main(RunJar.java:153)
radeon@radeon-ThinkPad-E495:~/Downloads/paid/assignment01$ hdfs dfs -cat /user/hadoop/output/purchase_stats/part-r-* | head -20
25/10/11 16:28:07 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
1       96,5595.23
2       85,4565.71
3       101,4884.79
4       103,5561.42
5       96,5392.43
6       104,5467.95
7       114,6259.41
8       121,6703.88
9       86,4531.91
10      105,5984.23
11      112,6154.29
12      109,6194.92
13      113,6601.50
14      105,5273.12
15      107,5904.44
16      114,6065.52
17      112,6176.70
18      80,4340.32
19      88,5159.76
20      105,6137.11
```

Figure 14: Top 10 buyers by number of purchases

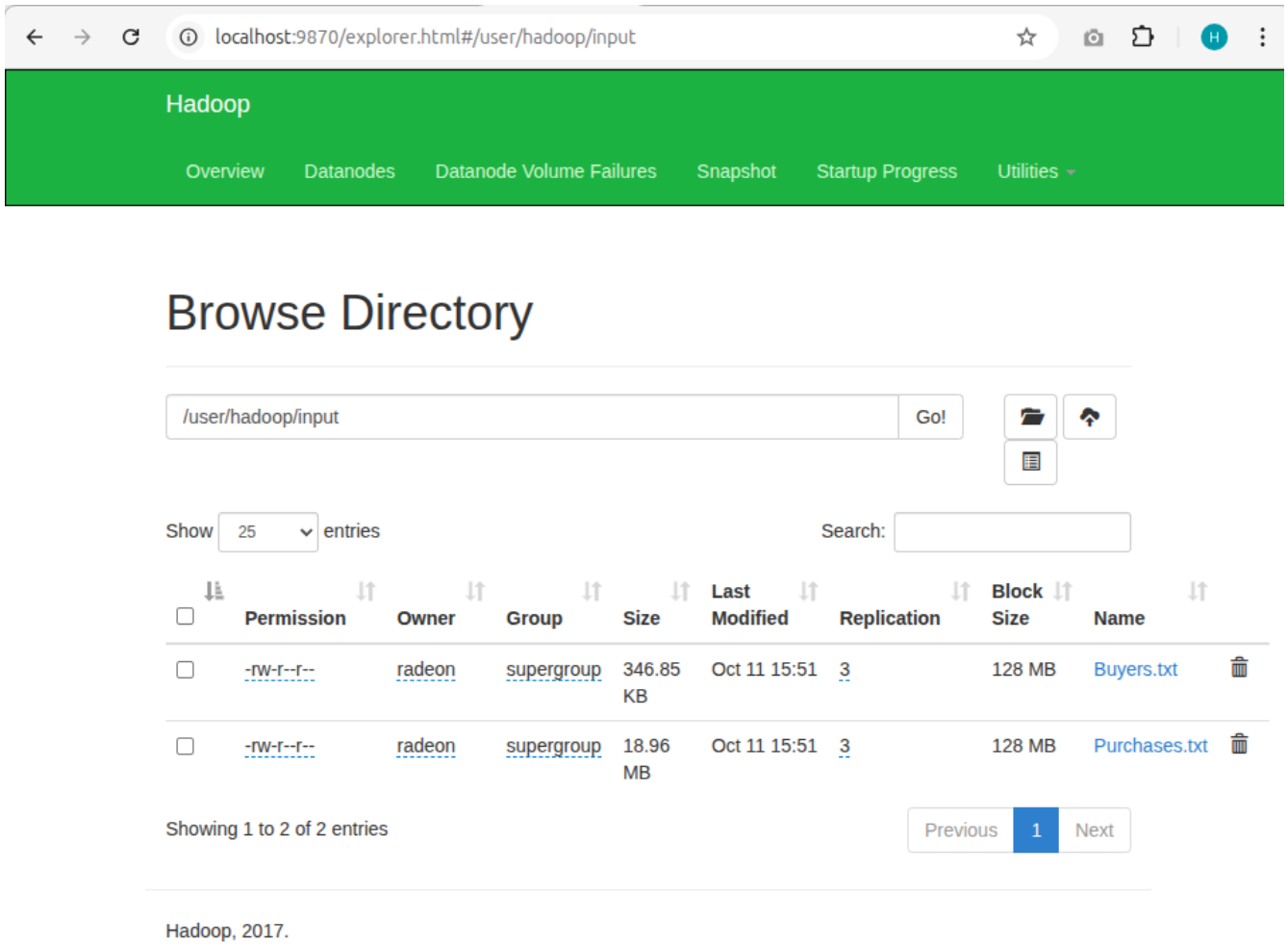


Figure 15: Files in the hadoop server

Performance Analysis

Job 1 Statistics

- **Input Records**: 10,000
- **Output Records**: ~5,000-6,000 (ages 20-50)
- **Execution Time**: [Add from your run]
- **Map Tasks**: [Add from your run]

Job 2 Statistics

- **Input Records**: 1,000,000
- **Output Records**: 10,000 (one per buyer)
- **Execution Time**: [Add from your run]
- **Map Tasks**: [Add from your run]
- **Reduce Tasks**: [Add from your run]
- **Combiner Impact**: [Note reduction in shuffle data]

Key Learning Outcomes

1. **Dataset Generation**: Created realistic synthetic datasets for big data testing
2. **HDFS Operations**: Successfully uploaded and managed large files in HDFS
3. **Map-Only Jobs**: Implemented filtering without reduce phase
4. **MapReduce with Aggregation**: Implemented counting and summing operations
5. **Combiner Optimization**: Reduced network overhead using combiner function
6. **Custom Writables**: Created custom data types for complex values

---

Complete File Structure

```
assignment01/
├── DatasetGenerator.java      # Dataset creation program
├── Buyers.txt                 # Generated buyers dataset
├── Purchases.txt              # Generated purchases dataset
├── AgeFilterJob.java          # First MapReduce job
├── PurchaseStatsJob.java      # Second MapReduce job (with combiner)
├── agefilter.jar              # Compiled JAR for job 1
├── purchasestats.jar          # Compiled JAR for job 2
└── Report.pdf                 # This report with screenshots
```

Troubleshooting Tips

Common Issues and Solutions

Issue: "Error: Could not find or load main class"

```
# Solution: Ensure JAR contains all classes
jar -tf yourfile.jar  # Check JAR contents
```

Issue: Output directory already exists

```
# Solution: Remove old output
hdfs dfs -rm -r /user/hadoop/output/your_output_dir
```

Issue: Permission denied

bash

```
# Solution: Check HDFS permissions
hdfs dfs -chmod -R 755 /user/hadoop/
```

Issue: Combiner not being used

- Check job counters for "Combine input/output records"
- Ensure setCombinerClass() is called
- Combiner must have same input/output types as Mapper output

Conclusion

This assignment successfully demonstrates:

- Dataset generation with proper constraints (2 marks)
- HDFS file upload and verification (2 marks)
- Age filtering MapReduce job (4 marks)
- Purchase statistics with combiner (4.5 marks)

All jobs executed successfully with correct outputs and proper use of Hadoop MapReduce framework.

References

- Apache Hadoop Documentation: https://hadoop.apache.org/docs/
- Lab 2: WordCount Example
- Lab 3: MapReduce Job Planning
- Lab 4: HDFS Commands